WEEK 1 DAY 5

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GROUP:S2

Pipeline Architecture & Experiment Planning

1. Pipeline Flowchart

Raw Data Input (V, I, P, T, History)

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Data Ingestion & Quality Audit

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Leakage-Safe Split (70/15/15 by Session)

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Feature Engineering (Power, Derivatives, Rolling Stats)

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Baseline Models (Dummy, Linear Regression)

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Advanced Models (Random Forest, XGBoost, ANN)

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Project-Specific Analysis (SoC Range Errors)

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Evaluation & Output (MAE, RMSE, R^2)

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Final Predictions:

- **SoC (%)**
- **Charging Time (minutes)**

2. **Model & Experiment Strategy**

- **Baselines:** Start simple with Linear Regression and Ridge Regression. These give us quick lower bounds and help check if the data behaves as expected.
- **Advanced Models:** Move to stronger methods like Random Forest and XGBoost, which can capture the non-linear charging behavior of batteries. If the dataset is large enough, we'll also test a neural network (ANN) to see if it adds value.
- **Project-Specific Analysis:** Beyond accuracy, we'll break down errors by charging phase and SoC ranges (Low: 0–30%, Mid: 30–80%, High: 80–100%). This helps us understand where models struggle most.
- **Evaluation Metrics:**
    - **MAE (Mean Absolute Error):** Average size of prediction errors.
    - **RMSE (Root Mean Squared Error):** Similar to MAE but penalizes large mistakes more.
    - **R^2 (Coefficient of Determination):** How much variance in the data the model explains.

Formulas:

$$MAE = \frac{1}{n}\sum_{i=1}^{n} | y_i - \hat{y}_i |$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)}$$

3. Repository & Directory Structure │

```
├── data/
│   ├── raw/
│   │   └── SOC_DATA_SET.zip
│   └── processed/
│       ├── X_train_scaled.csv
│       ├── X_val_scaled.csv
│       ├── X_test_scaled.csv
│       ├── y_train_soc.csv
│       ├── y_val_soc.csv
│       ├── y_test_soc.csv
│       ├── y_train_time.csv
│       ├── y_val_time.csv
│       ├── y_test_time.csv            │
```

```
├── notebooks/
│   ├── 01_eda_and_cleaning.ipynb
│   ├── 02_feature_engineering.ipynb
│   ├── 03_baseline_models.ipynb
│   ├── 04_advanced_models.ipynb
│   ├── 05_model_tuning.ipynb
│   └── 06_analysis_and_reporting.ipynb
│
├── models/
│   ├─ xgboost_soc.pkl
│   ├── xgboost_time.pkl
│   ├── random_forest_soc.pkl
│   ├── random_forest_time.pkl
│   ├── linear_regression_soc.pkl
│   └── linear_regression_time.pkl
```

4.Detailed Train/Validation/Test Split Information

4.1 Split Strategy Overview

The dataset is divided into three parts to ensure fair training, tuning, and evaluation:

Split	Percentage	Purpose	Number of Sessions	Number of Rows
Training Set	70%	Model training and learning	[To be filled]	[To be filled]
Validation Set	15%	Hyperparameter tuning and model choice	[To be filled]	[To be filled]
Test Set	15%	Final held-out evaluation	[To be filled]	[To be filled]

Here's a **shortened, page-ready version** of your Train/Validation/Test Split documentation. It condenses the details into 1–2 pages while keeping the structure professional and easy to paste into Word:

Train/Validation/Test Split Information

Split Strategy

The dataset is divided into three parts to ensure fair training, tuning, and evaluation:

Split	Percentage	Purpose
Training Set	70%	Model training and learning
Validation Set	15%	Hyperparameter tuning and model choice
Test Set	15%	Final held-out evaluation

Splitting is done by **session_id** to avoid leakage. Each charging session belongs to only one split, ensuring realistic performance estimates.

Split Specifications

Training Set (70%)

- Used to learn model patterns.
- Features: voltage, current, power, temperature.
- Targets: SoC (%) and charging time (minutes).
- StandardScaler fitted only on this set.

Validation Set (15%)

- Used for hyperparameter tuning and model selection.
- Same features and targets, scaled using training scaler.
- Not used for training.

Test Set (15%)

- Held-out set for final evaluation.
- Same features and targets, scaled using training scaler.
- Completely unseen during training and tuning.

Distribution Checks

To ensure fairness, distributions of SoC, charging time, voltage, current, and temperature are compared across splits. Acceptable differences:

- SoC mean < 5%
- SoC std dev < 10%
- Time mean < 10 min
- Time std dev < 15 min
- Voltage/current mean < 5%
- Temperature mean < 10%

Experiment Log Template

Experiments are tracked in a structured log file:

Exp ID	Model Name	Target	Status
EXP_01	Dummy Regressor	SoC	Baseline
EXP_02	Dummy Regressor	Time	Baseline
EXP_03	Linear Regression	SoC	Baseline
EXP_04	Linear Regression	Time	Baseline
EXP_05	Random Forest	SoC	Non-linear
EXP_06	Random Forest	Time	Non-linear
EXP_07	XGBoost	SoC	Primary
EXP_08	XGBoost	Time	Primary
EXP_09	XGBoost (Tuned)	SoC	Optimized
EXP_10	XGBoost (Tuned)	Time	Optimized
EXP_11	ANN	SoC	Optional
EXP_12	ANN	Time	Optional

Evaluation Metrics

Models are evaluated separately for both targets (SoC % and charging time in minutes):

- **MAE (Mean Absolute Error):** Average prediction error
- **RMSE (Root Mean Squared Error):** Penalizes large errors
- **R^2 (Coefficient of Determination):** Variance explained

Target values:

- SoC → MAE < 5%, RMSE < 8%, $R^2 > 0.80$
- Charging Time → MAE < 10 min, RMSE < 15 min, $R^2 > 0.70$